

# Introduction

The rise of connected devices (IoT) and the widespread adoption of distributed architectures such as Edge, Fog and Cloud are reshaping cybersecurity. By bringing processing closer to data sources, these models reduce latency, enhance resilience and enable adaptability. However, this technical shift also leads to fragmented controls, dispersed responsibilities and weaker traditional governance frameworks. Governance and auditability follow NIST and ISO/IEC standards: structured policy control and verifiable, traceable security actions.

**Research question:** How and to what extent does the growing adoption of distributed architectures (Edge, Fog, Cloud-IoT) create specific operational challenges for the effective cybersecurity management of modern IoT systems?

**Hypothesis:** While integrating distributed architectures into IoT ecosystems improves performance and resilience, it also significantly complicates cybersecurity management, challenging the effectiveness and relevance of traditional centralized security approaches.

Two mechanisms illustrate this complexity:

- the dispersion of control points undermines conventional authentication, segmentation and monitoring, reducing global visibility and complicating control orchestration;
- detection latency, exacerbated by the hardware limitations of peripheral nodes and the dynamic nature of distributed flows, hampers incident response and increases the risk of sustained compromise.

This review examines how recent literature addresses these dimensions, assessing conceptual rigour, empirical grounding, applicability, and unresolved gaps. The analysis draws on a selection of rigorous sources and is structured into two sections and an interim synthesis.

## Methodology

This review adopts an exploratory, inductive, and qualitative approach, based on the critical analysis of secondary sources from recent peer-reviewed publications.

Documents were selected for their direct relevance to the two focus areas (control governance and incident detection) and evaluated against four criteria: conceptual validity, empirical robustness, operational feasibility, and attention to distributed system constraints.

A triangulation of perspectives (centralised, decentralised, hybrid) enabled the cross-analysis of theoretical models and implementation limitations. Biases related to contextual variability or the field's rapid evolution were mitigated through methodological diversity and comparative examination of contrasting architectures and approaches.

## Dispersion of control points

In Edge-Fog-Cloud architectures, processing, storage and control are decentralized towards peripheral layers. This relocation reduces latency, improves resource distribution and enhances resilience to disruptions. However, it also fragments security governance.

Each entity (Fog node, Edge gateway or connected device) becomes a partial control and exposure point. Control dispersion undermines CIA principles, challenging unified oversight and access consistency across layers (NIST SP 800-207). The attack surface expands, control mechanisms are dispersed, and protections based on centralized oversight lose effectiveness. This setup complicates supervision and access consistency across heterogeneous environments.

Such context requires a rethinking of governance models and the design of mechanisms capable of maintaining secure control across distributed and heterogeneous decision points.

#### **a. centralized vs distributed governance**

Centralising control in Cloud creates a structural dependency, leading to latency in detection and response, and introducing a critical vulnerability that could compromise the entire system.

In response, studies propose hierarchical models. Masip-Bruin et al. (2016) and Velasquez et al. (2017) propose a hybrid Fog orchestration model, where management functions are split between centralized control at Cloud level and distributed decision-making at the Fog nodes. This decoupling reduces latency while preserving global coherence, even in heterogeneous environments.

Murad and Rahimi (2024) propose a decentralized approach. Their hierarchical peer-to-peer model, based on a ring structure and optimized local routing (LDEPTH), eliminates any explicit central authority. This design enhances node autonomy and fault tolerance, but raises concerns about synchronisation and consistency in distributed decision-making.

Two visions clash: vertical (Cloud and local orchestration) vs horizontal (no central control). Hybrid models remain unproven.

#### **b. Resilience and redundancy in distributed controls**

Murad and Rahimi (2024) propose a hierarchical peer-to-peer structure for Fog nodes, designed to maintain operational continuity in the event of compromise. Their model relies on redundant routing logic based on LDEPTH equations, removing the need for central supervision and enhancing the system's structural resilience.

However, robustness depends on several strong assumptions: a minimum level of trust between nodes, lightweight but secure encryption, and the local capacity to perform complex functions. In Fog environments constrained by energy, memory and bandwidth, these requirements significantly limit the model's portability.

Anthony (2023) highlights that lacking formal coordination in distributed systems leads to fragmented responsibilities and inconsistent policies. This undermines global oversight, especially in multi-actor environments without unified decision-making, and complicates secure policy synchronization, ultimately weakening operational resilience.

Literature shows tension: P2P offers resilience, but critics doubt its ability to ensure secure governance without strong oversight.

### **c. Dynamic orchestration and adaptive governance**

Several recent studies explore dynamic orchestration strategies aimed at real-time adjustment of security-related flows and responsibilities. These approaches rely on technologies such as SDN, NFV or AI. Kalyashina et al. (2024) advocate for the automatic reconfiguration of control functions based on load, detected threats or network topology.

Such mechanisms provide essential flexibility in distributed environments, but they rely on a central SDN controller, and that becomes a critical vulnerability. Despite its programmability, it reintroduces logical centralisation and exposes the control plane to targeted attacks. Implementation complexity, resource demands and lack of standards limit deployment.

Masip-Bruin et al. (2016) highlight this paradox: the apparent decoupling of control and data remains skewed by implicit recentralisation. In contrast, Szabo et al. (2025) examine distributed decision-making in dynamic, constrained environments. They describe models where autonomous entities act on partial context without central coordination. While improving fault tolerance and adaptability, this approach faces challenges on missing standards, synchronization complexity, global inconsistency risks, and limited empirical validation.

The literature contrasts two models: structured but fragile programmable central orchestration, and flexible but unstable emergent decentralized coordination. In both cases, adaptively managing control points remains an unresolved challenge.

## **Latency in security incident detection**

In Edge-Fog-Cloud environments, rapid incident detection is essential for responding before threats can spread. However, topological complexity, flow dispersion and device heterogeneity make such responsiveness uncertain.

Latency, the delay to identify suspicious behaviour, becomes a critical factor. It is exacerbated by fragmented observation points, data variability and the limited resources of Fog/Edge nodes, which hinder real-time embedded analysis.

This configuration weakens detection, delays response and raises risk of persistent compromise. Reducing latency requires moving beyond classical models to design mechanisms suited to distributed, constrained systems that often remain partially invisible to centralized supervision.

### **a. centralized vs embedded detection: an unstable compromise**

Most intrusion detection system (IDS/NIDS) architectures in IoT remain centralized, with systematic data transmission to Cloud. Bablu and Rashid (2020) argue that cloud-centric architectures are unfit for IoT latency constraints. Sending large data volumes to Cloud clogs

networks and delays critical responses. In contrast, offloading to Edge nodes reduces traffic and enables faster detection, especially in high-throughput environments.

In contrast, Cuadrillero Paredes et al. (2024) advocate for local deployment of detection capabilities at the Fog layer, using embedded behavioral analysis models. This approach helps reducing identification latency but shifts complexity to model training, supervision and synchronisation.

A comparative analysis of these strategies reveals a structural tension. Centralized model ensures overall consistency but sacrifices responsiveness. The Fog model improves detection speed but depends on limited local resources and requires technical oversight. Both studies agree on the existence of an unstable compromise between global coverage and local efficiency.

Masip-Bruin et al. (2016) propose a hybrid architecture: local detection coordinated through adaptive supervision based on event severity. This model aims to reconcile speed and consistency but remains conceptual, with no proven real-world implementation to date.

#### **b. Distributed Machine Learning and Deep Learning: effectiveness vs feasibility**

Diro and Chilamkurti (2018) show that pre-trained deep learning models deployed on Fog nodes can detect advanced attacks (DDoS, injection, man-in-the-middle) locally, with high accuracy and minimal latency. These results confirm the potential of embedded DL for decentralized detection without Cloud dependency.

However, practical feasibility remains limited. Local training is resource-intensive (CPU, memory, energy) and incompatible with edge constraints. Hardware heterogeneity undermines performance stability. Maintaining models requires a full pipeline difficult to scale: data collection, labelling, synchronisation, updates.

Kalyashina et al. (2024) propose federated learning to address this. Models are trained globally from local data without transferring raw inputs. This reduces bandwidth use and protects privacy, but remains experimental and does not solve synchronisation in unstable environments.

The contrast between promising lab results and the logistical barriers highlights a technical impasse. Performance is achievable, but structural viability remains out of reach.

#### **c. Proactive and preventive detection: anticipating attacks**

Masip-Bruin et al. (2016) and Gkonis et al. (2023) explore proactive detection in distributed architectures. These approaches aim to anticipate malicious behavior by analyzing weak signals and emerging anomalies. While promising, they rely on stable data flows and lack standardized formats, which limits their robustness in Fog and Edge contexts.

They rely on a mix of heuristics, historical profiling and statistical correlations. Their strength lies in generating probabilistic alerts before an attack materialises. However, their

effectiveness depends on data quality, network stability and the accuracy of the reference model.

Ahmad et al. (2023) highlight that real-time execution of deep learning models in Edge and Fog architectures is heavily constrained by latency, bandwidth, and limited local resources. They emphasize the need for model simplification, through pruning or quantization to enable deployment on resource-constrained edge nodes.

In response to this complexity, Diro and Chilamkurti (2018) favor a more operational approach: reactive, embedded detection that is not predictive but designed for efficient local execution.

Two logics are in tension: anticipation, often unstable in constrained environments, and embedded reactivity, more reliable but non-preventive. Neither successfully combines real-time execution, accuracy, and stable decision-making in distributed systems.

## Limitations

Recent literature on governance and detection in distributed IoT architectures highlights several structural shortcomings that continue to hinder real-world implementation.

A recurring issue lies in the overestimation of peripheral nodes capabilities. Many models assume advanced processing or routing that exceed the practical limits of energy-constrained, low-power devices. This gap between design and feasibility raises concerns about the actual deployability of these approaches.

Validation is another gap. Most proposals are tested in simulated settings or with synthetic datasets, offering limited real-world insight. As a result, key performance metrics such as resilience, detection speed and scalability remain largely unverified.

Coordination across layers (Edge, Fog and Cloud) also lacks formal definition. While various models refer to hierarchical or distributed schemes, they rarely specify how decisions are synchronised, alerts propagated or responsibilities allocated across nodes. This undermines system consistency.

Auditability is similarly underdeveloped. Distributed frameworks often fail to incorporate verifiable logging mechanisms, essential for forensic analysis and trust. Without robust, time-aligned audit trails, claims of resilience remain largely theoretical.

Fragmentation is further reinforced by the absence of interoperability standards. policies and data vary across platforms, limiting integration and obstructing collective oversight.

Lastly, many so-called decentralized models retain implicit central points of control, often via SDN controllers or cloud-based orchestrators. These elements reintroduce vulnerabilities that the architecture was meant to avoid.

Together, these limitations reveal a persistent gap between conceptual advances in distributed IoT security and the technical maturity required for their practical adoption.

# Perspectives

The literature reveals a persistent challenge: no current approach fully reconciles decentralisation, security, coordination and auditability within distributed IoT architectures. These approaches align with Zero Trust principles: no implicit trust, least privilege, and continuous verification (NIST 800-207). While peer-to-peer designs, software-based orchestrations and multi-agent systems offer promising directions, each introduces its own complexities in terms of stability, traceability or governance.

Three research priorities emerge. The first relates to adaptive orchestration, where security functions are dynamically distributed based on local load, bandwidth or energy. Lightweight embedded models could enable more responsive, context-aware decision-making.

The second concerns flexible control delegation. Rather than maintaining a strict binary between centralized and decentralized approaches, some proposals advocate for variable levels of authority depending on node role, exposure or risk profile. This federated logic requires formal frameworks, especially for identity and access.

The third area addresses distributed auditability. Ensuring traceability and non-repudiation without central authority calls for robust inter-node mechanisms, including shared ledgers, fault-tolerant consensus protocols or lightweight distributed ledger technologies.

In terms of detection, several avenues converge. Federated learning enables aggregation of local models without exposing raw data, but raises synchronisation and resilience challenges. Model simplification offers more practical deployment at the cost of reduced accuracy. Other approaches rely on local agent cooperation via trust-based rules. While promising, these models remain fragile without structural guarantees.

Taken together, these directions point to a shared goal: the design of embedded, adaptive and interoperable mechanisms capable to operate in constrained environments. The challenge is no longer simply to distribute control or minimise latency, but to do so with coherence, traceability and resilience.

# Conclusion

This review has highlighted key challenges in securing IoT ecosystems built on Edge-Fog-Cloud architectures. While distributing functions can reduce latency, enhance resilience, and localise processing, it also complicates governance, detection, and auditing.

The reviewed approaches reflect a shift from centralised control to more flexible but still unstable models, often unable to ensure local autonomy, global coherence, and traceability. This fragility stems from unrealistic assumptions, limited real-world validation, and poor interoperability, hindering adoption.

Emerging directions (adaptive orchestration, federated control, distributed auditing, federated learning, and multi-agent systems) point to contextual, modular, resource-aware

architectures. Yet these remain dependent on stronger formalisation, shared standards, and empirical testing.

Truly secure distributed IoT systems require moving beyond the binary of centralisation versus decentralisation, towards hybrid frameworks that combine operational efficiency, distributed governance, and verifiable auditability.

## References:

- Ahmad S, Shakeel I, Mehruz S and Ahmad J (2023) Deep learning models for cloud, edge, fog, and IoT computing paradigms: Survey, recent advances, and future directions. *Computer Science Review* 49, 100568. Available at: <https://doi.org/10.1016/j.cosrev.2023.100568> (Accessed: 2 June 2025)
- Anthony B (2023) A developed distributed ledger technology architectural layer framework for decentralized governance implementation in virtual enterprise. *Information Systems and e-Business Management* 21(2), pp 437-470. Available at: <https://doi.org/10.1007/s10257-023-00634-2> (Accessed: 3 June 2025)
- Bablu TA and Rashid MT (2020) Edge computing and its impact on real-time data processing for IoT-driven applications. In: *Proceedings of the International Conference on Computing Advancements (ICCA 2020)*. ACM, pp 1-6. Available at: <https://scipublication.com/index.php/JACS/article/view/117> (Accessed: 6 June 2025)
- Cuadrillero Paredes L, Fernandez-Maimo L, Llorente S, Perales-Gonzalez M, Gil D and Garcia-Gomez JM (2024) Advancing threat detection in fog computing: A comprehensive approach to real-time analysis and model generation. Available at: <https://ieeexplore.ieee.org/document/10710268> (Accessed: 5 June 2025)
- Diro AA and Chilamkurti N (2018) Distributed attack detection scheme using deep learning approach for Internet of Things. *Future Generation Computer Systems* 82, pp 761-768. Available at: <https://doi.org/10.1016/j.future.2017.08.043> (Accessed: 1 June 2025)
- Gkonis PK, Nomikos N, Trakadas P, Sarakis L, Xylouris G, Masip-Bruin X and Martrat J (2023) Leveraging network data analytics function and machine learning for data collection, resource optimization, security and privacy in 6G networks. *IEEE Access* 11, pp 36911-36923. Available at: <https://ieeexplore.ieee.org/abstract/document/10416824> (Accessed: 4 June 2025)
- International Organization for Standardization (ISO), 2013. *ISO/IEC 27001:2013 -Information technology - Security techniques -Information security management systems -Requirements*. Geneva: ISO.
- Kalyashina A, Orlova E, Pogosbekyan L and Slivina E (2024) Enhancing IoT systems through cloud-fog-edge architectures: Challenges and opportunities. *E3S Web of Conferences* 583, 06012. Available at: <https://doi.org/10.1051/e3sconf/202458306012> (Accessed: 2 June 2025)
- Masip-Bruin X, Marin-Tordera E, Garcia J, Rennings A, Verikoukis C and Becvar Z (2016) Foggy clouds and cloudy fogs: A real need for coordinated management of fog-to-cloud computing systems. *IEEE Wireless Communications* 23(5), pp 120-128. Available at: <https://ieeexplore.ieee.org/document/7721750> (Accessed: 30 May 2025)
- Murad SA and Rahimi N (2024) Secure and efficient hierarchical P2P fog architecture: A novel approach for IoT. *IEEE Internet of Things Journal* 11(10), pp 18796-18801. Available at: <https://doi.org/10.1109/JIOT.2024.3365071> (Accessed: 28 May 2025)

National Institute of Standards and Technology (NIST), 2018. Framework for improving critical infrastructure cybersecurity, Version 1.1. Gaithersburg, MD: U.S. Department of Commerce. Available at: <https://doi.org/10.6028/NIST.CSWP.04162018> [Accessed 21 May 2025].

National Institute of Standards and Technology (NIST), 2020. SP 800-207: Zero trust architecture. Gaithersburg, MD: U.S. Department of Commerce. Available at: <https://doi.org/10.6028/NIST.SP.800-207> [Accessed 21 May 2025].

Szabo R, Kovacs P and Toka L (2025) Overview and challenges of distributed decision making in Fog/Edge/Cloud computing. *Journal of Cloud Computing: Advances, Systems and Applications* 14(1), pp 1-15. Available at: <https://doi.org/10.1145/3719001> (Accessed: 1st June 2025)

Velasquez F, Moya F, Garrido D, Arrieta A and Macia-Pérez F (2017) SORTS: A framework for the classification of orchestration requirements and techniques for the cloud to the edge. In: Masip-Bruin X, Marín Tordera E and Garcia J (eds) *Orchestration from the Cloud to the Edge: A Complete Survey*. Springer, Chapter 2. (Accessed: 5 June 2025)